

What made Music popular?

A Retrospective of the last 20 Years

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Overview

1 Motivation

2 What we wanted to know

3 How we found out

- Data Sources
- Data Collection

“I think music in itself is healing.
It's an explosive expression of humanity.
It's something we are all touched by.
No matter what culture we're from,
everyone loves music.” – *Billy Joel*

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What we wanted to know I

Generally

How did various musical/lyrical elements vary in the most popular songs of the last 20 years?

What we wanted to know II

Concrete

- Rhythm
 - ▶ How did song duration change?
 - ▶ How did song tempo change?
- Chords
 - ▶ What chord-progressions repeat most frequently?
 - ▶ Do certain chord-progressions dominate specific genres?
 - ▶ What chords and chord types dominate?
 - ▶ How variant are the intervals between chords?
- Lyrics
 - ▶ Which words and combinations are the most frequent?
 - ▶ How did lyric polarity change?

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- **Main source:** *Billboard Year-End Hot 100 chart* [1] of 2005 up to 2024 with a (Title, Artist) table for each year.
- **Refined sources using songs from the main source:**
 - ▶ *Spotify API* [5] for song duration
 - ▶ *SongBPM* [4] webscraping for tempo
 - ▶ *Ultimate Guitar* [7] webscraping for chords
 - ▶ *LastFM API* [3] for “topTags” (genres)
 - ▶ *Genius API* [2] for lyrics and *TextBlob* [6] for analysis

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- Rhythm

- ① For each song retrieve duration [5] and tempo [4]
- ② Create (Title, Artist, Duration) and (Title, Artist, Tempo) tables.

- Chords

- ① Use chord-link finder to search for links on *Ultimate Guitar* [7] for each (Title, Artist) tuple.
- ② Run chord webscraper to scrape each chord for found song links.
- ③ Create (Title, Artist, Chords) tables.
- ④ Scrape genres from *LastFM API* [3] and create (Title, Artist, Top_Tags) table.

- Lyrics

- ① Use *Genius API* [2] to retrieve links to lyrics.
- ② Scrape all lyrics and create (Title, Artist, Lyrics_URL, Lyrics) tables.

References

- [1] *Billboard Year-End Hot 100 chart*. URL: https://en.wikipedia.org/wiki/Category:Lists_of_Billboard_Year-End_Hot_100_singles.
- [2] *Genius API*. URL: <https://api.genius.com>.
- [3] *LastFM API*. URL: <https://www.last.fm/api>.
- [4] *SongBPM*. URL: <https://songbpm.com/>.
- [5] *Spotify API*. URL: <https://developer.spotify.com/documentation/web-api/reference/search>.
- [6] *TextBlob*. URL: <https://textblob.readthedocs.io/en/dev/index.html>.
- [7] *Ultimate Guitar*. URL: <https://www.ultimate-guitar.com/>.